

UCE803 : IRRIGATION ENGINEERING				
	L	T	P	Cr
	3	1	0	3.5
Course Objective: To expose students to various methods of irrigation and design of hydraulic structures				
Introduction: Introduction, types and methods of irrigation, concept of smart irrigation systems				
Water Requirement of Crops: Soil-moisture-irrigation relationship, depth and frequency of irrigation, irrigation efficiencies, consumptive use and its determination, duty and delta relationship, factors affecting duty, crop seasons.				
Canal Irrigation: Canal irrigation system, canal alignment, canal losses, estimation of design discharge of a canal, design of stable channels by Lacey's and Kennedy's theory, Water logging, design of tile drains.				
Canal Head works: Layout and component parts of a diversion headwork scheme, design of weir/barrage as per IS specifications				
Design of Impermeable floor: Causes of failure of hydraulic structures founded on previous foundations, Bligh's creep theory and Khosla seepage theory, hydraulic jump and its applications in the design of hydraulic structures, design of a canal head regulator.				
Canal Regulation Works: Canal falls, necessity, location, and types of falls, design of a vertical drop fall and a glacis fall, roughening measures for energy dissipation, cross regulators and distributary's head regulators, canal escape and canal outlets				
Cross Drainage (CD) Works: Need, types, selection of suitable CD work, design of CD works				
Course Learning Objectives (CLO)				
Upon completion of this course, the students will be able to:				
1. Work out the water requirement of crops. 2. Solve problems related to different structures needed for diverting and regulating irrigation water 3. Describe and define different types of cross drainage works				
Text Books				
1. S.K. Garg, <i>Irrigation Engineering and Hydraulic Structures</i>, Khanna Publishers, New Delhi (2015). 2. P.N.Modi, <i>Irrigation Water Resources and Water Power Engineering</i>, Standard Book House, New Delhi(2014) 3. Bharat Singh, <i>Fundamentals of Irrigation Engineering</i>, Nem Chand, Roorkee (2005)				

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations)	25